Maansa Krovvidi

mkrovvid@ucsc.edu | Linkedin
Santa Cruz, California - 95060, United States

Research Overview

My research interests focus on the Internet of Things (IoT), wearable sensing, sensor fusion, and cyber-physical systems. I am particularly interested in developing novel sensing techniques and end-to-end systems that address real-world challenges in agriculture, environmental monitoring, healthcare, and wellness tracking. My previous work has explored applications such as activity recognition, posture correction, and resource intake tracking. Moving forward, my goal is to design sustainable, energy-efficient sensing solutions that leverage non-traditional energy sources and operate reliably in real-world environments.

Education

- **University of California, Santa Cruz** *Sept 2025 - May 2030*
  PhD in Electrical and Computer Engineering - Advisor: Prof. Colleen Josephson California, USA
  - Research Interests: UWB sensing, Internet of things(IoT), Energy-efficient systems, Wireless sensor networks
  - Scholarships: AES Scholarship (Summer '26), UC Regents Scholarship recipient (Fall '25, Winter '26).
  - GPA: 4.00
- **Carnegie Mellon University** *Aug 2022 - May 2024*
  Master of Science in Mobile and IoT Engineering (MSMITE) - Advisor: Prof. Quinn Jacobson California, USA
  - Thesis: Activity Detection using RF and Sensor Data Fusion
  - GPA: 3.76
- **Chaitanya Bharathi Institute of Technology** *Aug 2017 - Aug 2021*
  Bachelor of Engineering in Electronics and Communication Engineering - Advisor: Prof. Sai Krishna Kondoju Hyderabad, India
  - Thesis: Performance Analysis of ZF-SIC and MMSE Equalizers for MIMO Systems in the Presence of Alpha(α)-Mu(μ) Fading Channel

Research Experience

- **jLab in Smart Sensing** *March 2026 - Ongoing*
  PhD Student Santa Cruz, CA, USA
  - Investigating low-power UWB sensing and passive backscatter communication for depth-resolved soil compaction profiling in agricultural soils.
  - Trained and validated machine-learning models to estimate soil bulk density from radar data, attaining 0.06 g/cm³ RMSE in intra-domain testing.
- **Bosch & Carnegie Mellon University** *Sept 2023 - Aug 2024*
  Wireless AI Student Researcher Sunnyvale, CA, USA
  - Enhanced a multimodal transformer by incorporating new features for precisely classifying 14 different human activities, achieving a 30% reduction in activity misprediction by leveraging both IMU and RF data.
  - Leveraged existing RF data from BLE protocols for fine-grain relative positioning to improve the precision of human biomechanical motion tracking.

Experience

- **Zappix** *Feb 2025 - June 2025*
  R&D Intern Boston, MA, USA
  - Developed a Retrieval-Augmented Generation (RAG) chatbot using LangChain, integrating web-scraped data, ChromaDB for vector search, and Streamlit for a functional UI.
  - Designed and integrated advanced prompt engineering techniques with context-aware retrieval, enhancing the chatbot's ability to generate accurate and coherent responses across diverse queries.
- **Rite Aid** *Aug 2024 - Feb 2025*
  Data Analyst Remote, USA
  - Worked on a custom AI copilot to enhance response accuracy by analyzing security risks in healthcare, using Anomali ThreatStream to identify compromised IPs.
- **Qualcomm** *May 2023 - Aug 2023*
  IoT Systems Intern San Diego, CA, USA
  - Designed an adaptive flexible scheduler to optimize process allocation on Qualcomm's AI and 5G-enabled RB5 platform to maximize per-core utilization while meeting power and 5ms latency requirements for an IoT-automated smart grid system.
  - Improved the efficiency of resource allocation across three gold cores by implementing dynamic scheduling while taking the per-core DMIPs into account, leaving a 25% margin in RTLinux.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[S.1] **Maansa Krovvidi**, Jake Lee, David Glover, Eric Vetha, Colleen Josephson, "GOPHERS: Low-power, Non-invasive Soil Compaction Sensing Using Commodity Radar," In Proceedings of the 28th International Workshop on Mobile Computing Systems and Applications (HotMobile '27). Association for Computing Machinery, Arizona, AZ, USA.

[C.1] **Maansa Krovvidi**, Zhiyi Shi, Sushanta Rakshit, Anagha Ravi Shankara, Vivek Jain, Quinn Jacobson, "Activity Recognition using RF and IMU Sensor Data Fusion," In Proceedings of the 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys '25).Association for Computing Machinery, New York, NY, USA, 104–109.

[T.1] **Maansa Krovvidi**, "Activity Detection Using RF and Sensor Data Fusion," M.S. thesis, Dept. of Information Networking Institute, Carnegie Mellon University, Mountain View, CA, 2024.

ACADEMIC PROJECTS

- **Wearable Sensor-Based Meal Tracker** *Jan 2024 - May 2024*
  Carnegie Mellon University Mountain View, CA, USA
  - Modified the Movesense device firmware to interpret hand gesture data for monitoring drinking, snacking, and other daily activities by utilizing Nordic Semiconductor's nRF52832 controller.
- **Practicum Project: Human Activity Recognition through Multi-Sensor Data Fusion** *Sept 2023 - Dec 2023*
  Carnegie Mellon University Mountain View, CA, USA, USA
  - Developed a Proof of Concept (PoC) for detecting 7 physical activities using 4 on-body BLE nodes by integrating additional sensing modalities such as RSSI and IMU.
  - Demonstrated that integration of additional sensing modalities resulted in improved performance in complex movements such as squats (100% accuracy) and static poses like Warrior Pose (100% accuracy).
- **PostURight: IoT Wellness App with Posture Monitoring** *Jan 2023 - May 2023*
  Carnegie Mellon University Pittsburgh, PA, USA
  - Created a hybrid IoT application to monitor back posture in real-time, using wearable sensors and interactive UI prompts developed with Flutter and Figma to enhance human physical well-being.
  - Conducted a comprehensive survey with 63 participants to evaluate security and usability risks linked to the wearable sensor, guiding personalized modifications to enhance application security via the Firebase database.
- **Smart Hand Gesture Recognition using Vision Transformer** *Sept 2022 - Dec 2022*
  Carnegie Mellon University Pittsburgh, PA, USA
  - Devised a robust ML algorithm for sign language-to-text translation using a web camera, fostering communication for speech-impaired individuals. Analyzed model accuracy for both the vision transformer and VGG-18 CNN models, achieving a test accuracy of 98.45% with a best cross-entropy loss of 0.037.

SKILLS

- **Programming Languages:** C/C++, Python, R, MATLAB, LATEX, Java, XML, MATLAB
- **Software and Analytical tools:** Scikit-Learn, NumPy, Pandas, PyTorch, TensorFlow, Matplotlib, Wireshark
- **App Development:** Figma, Flutter, Kotlin
- **IoT systems and protocols:** Linux, Windows, RTOS, Arduino, BLE, Firebase database, TCP/IP, BGP, OSPF

HONORS AND AWARDS

- **Best Team Project Award Winner** *April 2023*
  School of Computer Science, Carnegie Mellon University
  - Received the best project award for the project titled "PostuRight" in the course Mobile and IoT Computing Services (Spring 2023) conducted by Prof. Norman Sadeh.

TEACHING & MENTORSHIP EXPERIENCE

- **Mentor** *Sept 2024 - Present*
  INI Mentorship program, Carnegie Mellon University
  - Mentoring three students in the CMU Universe network, providing one-on-one guidance on course selection, networking through conferences, and interview preparation.
- **Teaching Assistant** *Jan 2023 - May 2023*
  Information Networking Institute, Carnegie Mellon University
  - Mentored four teams on customer preference modeling and acted as a liaison between the professor and students to clarify course material.
  - Offered office hours for one-on-one student support, addressing individual queries, guiding project work, and facilitating a deeper understanding of topics discussed in class.

Community Engagement and Outreach

- **Judge** *Nov 2025*
  Reboot The Earth Hackathon, UC Santa Cruz
  - Served as a first-round judge at the hackathon, evaluating 5 startup pitches against defined scoring criteria to help select finalists.

Professional Memberships

- **Women in Cybersecurity**, Membership ID: 71350942 *March 2023 - Present*
- **AnitaB.org**, Membership ID: maansa.madhuri *Sept 2023 - Present*

Certifications

- **Cybersecurity Essentials, Cisco Networking Academy** *Aug 2021*
- **Data Science: Visualization, HarvardX** *June 2020*
- **Data Science: R, HarvardX** *May 2020*
- **Machine Learning, Stanford University** *May 2020*

Hobbies

- Music: Indian Classical, Baking, Hiking, Photography: Landscape and Macro, Sports: Badminton